

# SADHANA JAMKATEL

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## EDUCATION

### North Central College

*Bachelor of Science in Computer Science*

May 2027

GPA: 3.87/4.0

**Relevant Coursework:** Object-Oriented Programming, Data Structures, Algorithms, Discrete Maths, Calculus 1&2, Operating Systems, Computer Organization, Networking, Databases, Data Science, Breakthrough Tech Machine Learning Course

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, JS/TS, C/C++, Java, SQL, HTML/CSS, React.js, Next.js, Node.js, Express, Flask

**Backend, Cloud & AI:** AWS (S3, EC2, CloudFront), Docker, PostgreSQL, MongoDB, Git, LangChain, scikit-learn, OpenCV

## EXPERIENCE

### Aeropay

Chicago, IL

*Software Engineering Intern (Full Stack) - Pay Team*

Jun 2026 - August 2026

- Engineered high-throughput admin API endpoints and refactored state management logic, eliminating operational bottlenecks and slashing merchant account configuration SLAs from **24 hours to seconds**.
- Embedded a generative AI agent (Claude API / AWS Bedrock) into the Enterprise Operations Platform to automate risk mitigations via REST endpoints - condensing multi-step manual workflows and **cutting response time by ~90%**.
- Designed stateful risk-mitigation processes with permission gates, confirmation dialogs, and real-time Slack/Event-driven audit logging, achieving **100% data traceability** for administrative actions.
- Redesigned banking microservices to support concurrent ACH and Real-Time Payment (RTP) capabilities via capability routing tables, authoring OpenAPI specs to achieve **100% read/write path parity**.
- Resolved complex backend edge cases across microservices - redesigning configuration update logic, resolving search pagination defects, and **fixing 500 server crashes in production**.

### Aeropay

Chicago, IL

*DevOps Intern*

Jun 2025 - August 2025

- Accelerated engineering velocity by building a centralized, full-stack technical documentation infrastructure using Docusaurus and GitHub Actions CI/CD, reducing information lookup times by 98% for 20+ engineers daily
- Built automated CI/CD documentation pipelines that automatically generated and published up-to-date user manuals for internal cloud infrastructure and security CLI tools

## PROJECTS

### Care Equity | Full-Stack Healthcare Bias Tracker (Hackathon Code2040)

- Built a Next.js, Express, and MongoDB analytics app with Recharts visualizations to surface maternal health disparities (3.5x mortality gap), ensuring **100% patient privacy and data security**.
- Designed a same-origin Express API proxy that eliminated **100% of mobile CORS errors**, cutting search lookup latency by ~40% and maintaining ~99.9% service uptime across Vercel and Render.

### Predictive Maintenance ML System for Fleet Engine Monitoring

- Built a Random Forest model analyzing 21 sensor streams from the NASA Turbofan dataset, utilizing custom feature engineering and hyperparameter tuning to achieve **96% accuracy** with a **4.28% false alarm rate**.
- Designed a React/TypeScript monitoring dashboard with real-time risk alerts across 100 fleet engines, reducing engine downtime by **95%** and delivering **\$750K+ in projected annual cost savings**.

## LEADERSHIP AND INVOLVEMENT

### Society of Hispanic and Professional Engineers - Vice President

August 2023 – Present

- Orchestrated technical workshops, hackathons, and guest lectures, promoting hands-on learning and bridging the gap between academia and industry, resulting in a **50% increase** in internship applications/success rate

### Resident Assistant

May 2024 – Present

- Mentored 40+ residents and organized 15+ semesterly community and professional development events, driving a **95% increase in resident participation** while fostering an inclusive, supportive living environment.

**Additional Involvement:** SHPE Vice President 2026, CodePath, ColorStack, Rewriting the Code, Kamala Scholar (GHC 2026)